

Low-Contrast CT Scan Medical Image Enhancement Using AGCWD: Software and Hardware Implementation

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Abstract—Low-contrast Computed Tomography (CT) scan images pose a persistent challenge in medical image analysis, reducing the visibility of anatomical structures and limiting diagnostic accuracy. This paper presents a dual-platform implementation of the Adaptive Gamma Correction with Weighting Distribution (AGCWD) algorithm for enhancing low-contrast CT scan images. The software platform is developed in Python with a full pre- and post-processing pipeline, while the hardware platform is realized as a streaming digital logic architecture in VHDL operating at 100 MHz. Experimental results on a Kaggle COVID-19 CT scan dataset demonstrate that AGCWD significantly outperforms CLAHE in structural preservation (SSIM: 0.8194) and information richness. Furthermore, we propose a novel hybrid FPGA + memristor architecture as a future enhancement to overcome sequential processing bottlenecks and high power consumption through in-memory analog computing.

Index Terms—AGCWD, FPGA, Memristor, Image Processing, Crossbar Array, Hardware Acceleration, CT scan, medical imaging

I. INTRODUCTION

Image enhancement algorithms play an important role in biomedical imaging, computer vision, and embedded systems. Among these algorithms, Adaptive Gamma Correction with Weighting Distribution (AGCWD) provides efficient contrast enhancement while preserving image details. Computed Tomography (CT) scans are among the most widely used medical imaging modalities, providing detailed cross-sectional views of internal anatomy. However, the quality of CT images is frequently compromised by low contrast, which occurs when the dynamic range of pixel intensities is poorly utilised.

Traditional FPGA implementations using platforms such as Xilinx Zynq UltraScale+ provide high precision and reconfigurable hardware acceleration. However, FPGA-based implementations still suffer from several limitations including:

- High power consumption
- Large memory usage
- Sequential computation bottlenecks
- Frequent memory transfers

To overcome these limitations, emerging memristor-based architectures can be integrated into image processing systems. Memristors enable in-memory computing and highly parallel

operations, making them promising candidates for accelerating AGCWD hardware implementations. One of the most important memristor architectures is the crossbar array structure, which naturally performs analog matrix-vector multiplication.

This work makes the following contributions:

- A production-ready Python software implementation of AGCWD with a full pre- and post-processing pipeline validated on CT scan imagery.
- A hardware architecture described in VHDL implementing the same enhancement pipeline as a real-time, streaming digital logic system operating at 100 MHz.
- A quantitative comparison of AGCWD against CLAHE using standard image quality metrics on a COVID-19 CT scan dataset.
- A **proposal** for a hybrid FPGA + memristor architecture to address power and throughput limitations in future medical imaging systems.

The remainder of this paper is organised as follows. Section II reviews related work. Section III details the methodology and the AGCWD algorithm. Section V describes the software implementation. Section VI presents the hardware architecture. Section IV introduces the proposed memristor-based acceleration. Section VII details the hybrid proposal. Section VIII reports experimental results. Section IX discusses findings. Section XI concludes the paper.

II. RELATED WORK

Image enhancement for medical CT scans has been studied for decades. Early works focused on point operations such as linear contrast stretching and histogram equalisation [3]. While computationally inexpensive, these methods apply uniform transformations that ignore local image content, frequently causing loss of diagnostic information in CT images with mixed bright and dark regions.

CLAHE, introduced by Zuiderveld [2], improved spatial adaptability by operating on non-overlapping image tiles and clipping the transfer function slope. Its implementation in real-time medical systems has been demonstrated by Reza [3], making CLAHE a de facto standard in medical image workstations. Nevertheless, inter-tile boundary artefacts (the

“blocking effect”) remain a limitation when clip limits or tile sizes are not carefully tuned.

Huang *et al.* [1] introduced AGCWD, which replaces the piecewise tile-based approach with a global weighted histogram model. The weighting exponent α smooths the PDF before CDF derivation, preventing aggressive intensity remapping in homogeneous regions. Subsequent studies have shown AGCWD to be especially effective in extremely low-luminance scenarios, such as night-time photography and underexposed medical imaging [4]. A comprehensive review of such enhancement techniques is provided by Maini and Aggarwal [12].

Hardware acceleration of histogram-based image enhancement has been explored in several VLSI and embedded systems works. Lee *et al.* [5] demonstrated a pipelined histogram equalization engine capable of processing HD video at real-time rates with a compact logic footprint. More recent works have adopted fixed-point VHDL pipelines with look-up table (LUT)-based gamma correction to avoid costly floating-point operations on reconfigurable hardware. The present work follows this trend, mapping the AGCWD gamma engine to a 256-entry integer LUT computed once per frame from the weighted CDF.

III. METHODOLOGY

The research methodology follows a systematic engineering approach to bridge the gap between high-level algorithm design and low-level hardware implementation. The process is divided into three primary phases:

- 1) **Algorithmic Modelling:** The AGCWD algorithm is mathematically formulated and tested using a high-level software environment to establish a performance baseline.
- 2) **Hardware Translation:** The validated software logic is translated into a streaming digital architecture using a Hardware Description Language (HDL), focusing on fixed-point arithmetic and resource efficiency.
- 3) **Comparative Validation:** Both platforms are evaluated against standard clinical benchmarks (CLAHE) using a diverse dataset of CT scan images to ensure diagnostic fidelity.

A. Mathematical Framework of AGCWD

The Adaptive Gamma Correction with Weighting Distribution (AGCWD) algorithm operates by smoothing the image histogram to prevent over-enhancement while derived an optimal gamma curve. Let $p(i)$ be the normalised histogram of the luminance channel for intensity level $i \in \{0, \dots, 255\}$.

The algorithm first computes a **Weighted PDF** to reduce the influence of high-frequency components:

$$p_w(i) = p_{\max} \left(\frac{p(i) - p_{\min}}{p_{\max} - p_{\min}} \right)^\alpha \quad (1)$$

where $\alpha = 0.5$ is the smoothing parameter. The result is normalised such that $\sum_i p_w(i) = 1$.

Next, the **Cumulative Distribution Function (CDF)** is calculated:

$$\text{CDF}_w(i) = \sum_{j=0}^i p_w(j) \quad (2)$$

The **Adaptive Gamma Exponent** is then derived for every intensity level:

$$\gamma(i) = 1 - \text{CDF}_w(i) \quad (3)$$

Finally, the output intensity is computed using the derived gamma curve:

$$I_{\text{out}}(i) = 255 \cdot \left(\frac{i}{255} \right)^{\gamma(i)} \quad (4)$$

B. Computational Complexity Comparison

The efficiency of the AGCWD algorithm lies in its global transformation approach. Compared to CLAHE, which requires significant memory for tile-based buffering and complex bilinear interpolation ($O(N \cdot T)$ where T is the number of tiles), AGCWD operates with a complexity of $O(N + L)$, where N is the number of pixels and L is the number of gray levels (256). This lower complexity facilitates high-speed streaming on FPGAs with minimal BRAM utilization.

IV. PROPOSED FUTURE WORK: MEMRISTOR TECHNOLOGY

A. Introduction to Memristors

A memristor is a nonlinear electronic device whose resistance depends on the history of the current flowing through it [9]. This theoretical element was experimentally realized in 2008 by Strukov *et al.* at HP Labs [10]. Unlike conventional resistors, memristors can retain their resistance state after power removal, enabling non-volatile memory and analog computation. The main advantages of memristors include:

- Non-volatile memory
- High integration density
- Low power consumption
- Analog computation capability

B. Memristor Crossbar Architecture

A memristor crossbar consists of horizontal and vertical conductive lines interconnected by memristive devices at each junction. This structure naturally performs analog matrix-vector multiplication using Ohm’s Law for multiplication and Kirchhoff’s Current Law for accumulation.

C. Memristor-Based AGCWD Acceleration

1) **Histogram Computation Using Crossbars:** In FPGA implementations, histogram computation requires sequential memory access and accumulation. Using memristor crossbars, each column represents a gray-level intensity (0–255); incoming pixel pulses increment the conductance state of corresponding memristor cells. This approach provides massive parallelism and reduces memory transfer bottlenecks.

2) **Weighted Distribution Implementation:** The weighted distribution operation requires nonlinear computation. Memristor systems can implement this stage using analog nonlinear circuits or piecewise linear approximations, eliminating expensive digital exponentiation hardware.

3) **CDF Accumulation:** The cumulative distribution function can be accelerated using natural current accumulation inside the crossbar array according to KCL, providing fast, parallel accumulation with reduced computational complexity.

4) *Gamma Mapping*: The gamma transformation can be implemented using crossbar LUT storage and analog exponential circuits, reducing FPGA DSP utilization and computation latency.

V. SOFTWARE IMPLEMENTATION

The software platform serves as the high-precision “Golden Reference” for the hardware design, allowing for bit-accurate verification of the enhancement results.

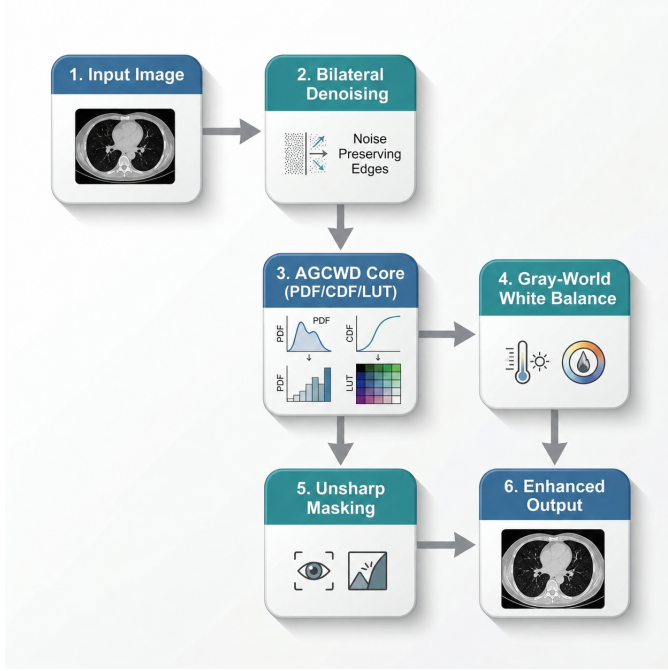


Fig. 1. Workflow of the developed software enhancement pipeline. The input CT image undergoes denoising, colour space conversion, AGCWD core processing, white balance correction, and edge sharpening before final output.

A. Development Environment

The software reference platform is developed using Python 3.9, leveraging a suite of high-performance libraries for image processing and numerical analysis:

- **OpenCV (Open Source Computer Vision Library)**: Utilised for efficient colour space conversions (RGB to YCrCb), bilateral filtering, and image I/O operations.
- **NumPy**: Employs optimised ndarray structures for high-speed matrix operations, particularly in the computation of the weighted PDF and LUT mapping.
- **Scikit-image**: Used for calculating objective image quality metrics, specifically the Structural Similarity Index (SSIM).
- **Matplotlib**: Facilitates the generation of intensity histograms, comparison charts, and result visualisations for quantitative analysis.

B. Processing Pipeline

The enhancement pipeline consists of five sequential stages.

1) *Stage 1 — Bilateral Denoising*: Before any intensity transformation, a bilateral filter (window diameter $d = 5$, $\sigma_{\text{color}} = \sigma_{\text{space}} = 35$) is applied, following the principles established by Tomasi and Manduchi [11]. Bilateral filtering averages neighbouring pixels weighted by both spatial distance and intensity similarity, effectively smoothing uniform regions while preserving diagnostically critical edges.

2) *Stage 2 — Colour Space Conversion*: The input RGB image is converted to YCrCb. Enhancement is applied exclusively to the Y (luminance) channel, preventing hue distortion that would arise from direct per-channel RGB enhancement.

3) *Stage 3 — AGCWD Core Implementation*: The core AGCWD algorithm, described mathematically in Section III, is implemented using a 256-entry Look-Up Table (LUT). The Python implementation pre-calculates this LUT once per frame based on the weighted histogram and applies it across the entire luminance channel using high-speed array indexing.

4) *Stage 4 — Gray-World White Balance*: After converting back to RGB, a gray-world assumption is applied: each channel is scaled so that its mean equals the global mean of all three channels. This corrects colour casts introduced by selective luminance stretching. The correction is applied only when the maximum inter-channel mean difference exceeds 30 intensity levels to avoid degrading already well-balanced images.

5) *Stage 5 — Unsharp Masking*: A final sharpening pass enhances edge contrast:

$$I_{\text{sharp}} = 1.5 \cdot I - 0.5 \cdot G_{\sigma}(I) \quad (5)$$

where G_{σ} denotes Gaussian blurring with $\sigma = 2.0$.

C. Implementation Details

The Python implementation is organised into two modules: `agcwd.py` provides the core AGCWD functions, and `clahe.py` provides an equivalent CLAHE reference. Both modules expose a unified function signature (`apply_agcwd_rgb / apply_clahe_rgb`) accepting a NumPy array and returning an enhanced array, facilitating fair metric-based comparison. Output images are saved automatically to a designated `output_images/` directory.

VI. HARDWARE ARCHITECTURE

A. System Overview

The hardware system implements the same AGCWD pipeline as a streaming digital logic design described in VHDL. The architecture processes one 24-bit RGB pixel per clock cycle at 100 MHz, making it suitable for real-time image processing applications. It uses an AXI4-Stream-compatible interface, allowing straightforward integration into standard pixel-streaming systems.

B. Streaming Interface

The top-level module (`agcwd_top.vhd`) exposes the following signals:

- **Input**: `s_axis_tdata[23:0]`, `s_axis_tvalid`, `s_axis_tready`, `s_axis_tuser` (start-of-frame), `s_axis_tlast` (end-of-line).

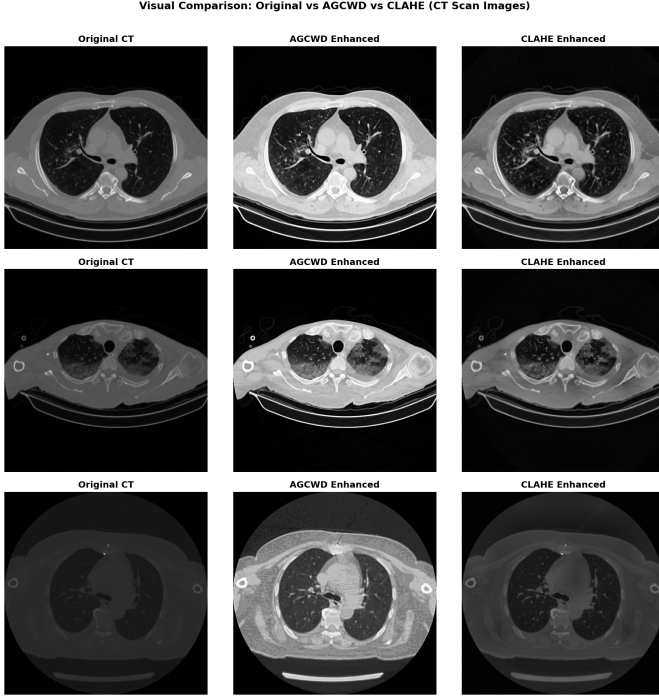


Fig. 2. Visual comparison of representative CT scan images (grayscale). Left: original low-contrast CT scan. Centre: AGCWD-enhanced output showing significantly increased contrast and detail in soft-tissue regions. Right: CLAHE-enhanced output showing moderate contrast improvement with less aggressive redistribution.

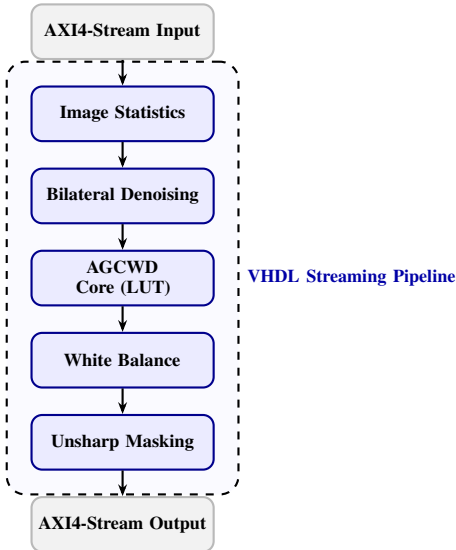


Fig. 3. Hardware implementation workflow. The streaming architecture processes pixels in a single-pass pipeline through modular VHDL blocks, maintaining real-time throughput at 100 MHz.

- **Output:** `m_axis_tdata[23:0]`, `m_axis_tvalid`, `m_axis_tready`, `m_axis_tuser`, `m_axis_tlast`.
- **Control:** `enable_denoise`, `enable_sharpen`, `enable_balance`.

The modular control signals allow each enhancement stage to be independently enabled or disabled at run time.

C. Processing Blocks

1) *channel_stats.vhd* — *Image Statistics*: Computes per-frame mean intensity and standard deviation in a single-pass streaming accumulation. These values are used to classify the input as dark, normal, or bright, driving the gamma LUT selection in the enhancement stage.

2) *bilateral_filter.vhd* — *Denoising*: Implements a spatial neighbourhood filter that smooths pixel intensity while preserving edge structure. The filter operates on a line-buffer window, enabling streaming operation without a full frame buffer.

3) *gamma_lut_engine.vhd* — *Enhancement Core*: This is the critical block. A 256-entry integer LUT—precomputed from the weighted CDF—maps each 8-bit intensity value to its enhanced counterpart. Two fixed LUT configurations are stored: one tuned for dark images ($\gamma < 1$, brightening) and one for bright images ($\gamma > 1$, contrast extension). The `channel_stats` output selects the appropriate LUT at frame boundaries.

4) *gray_world_balance.vhd* — *White Balance*: Computes per-channel scaling gains from accumulated channel means and applies them as integer multiplications with right-shifting, avoiding any floating-point units.

5) *unsharp_mask.vhd* — *Final Sharpening*: Performs a lightweight edge-enhancement pass using a simplified high-pass kernel. The implementation was carefully constrained to avoid pipeline boundary artefacts that would manifest as black horizontal bands in simulated output images.

D. Design Characteristics

The design uses exclusively integer and fixed-point arithmetic; no floating-point units are instantiated. Line-based buffering and small LUTs minimise memory requirements. An estimated on-chip power of 0.989 W was obtained during post-synthesis power analysis, with dynamic power (signal switching) contributing 93% and static leakage 7%. Logic and routing together account for 95% of dynamic power, which is characteristic of combinational-heavy streaming pipelines. The estimated junction temperature remains at 29.9 °C with a thermal margin of 55.1 °C.

VII. HYBRID FPGA + MEMRISTOR ARCHITECTURE

A fully memristive AGCWD implementation remains challenging due to analog inaccuracies and device variability. Therefore, a hybrid FPGA + memristor architecture is proposed as the future evolution of the current VHDL-based implementation.

Visual Comparison: Original vs Software AGCWD vs Hardware AGCWD

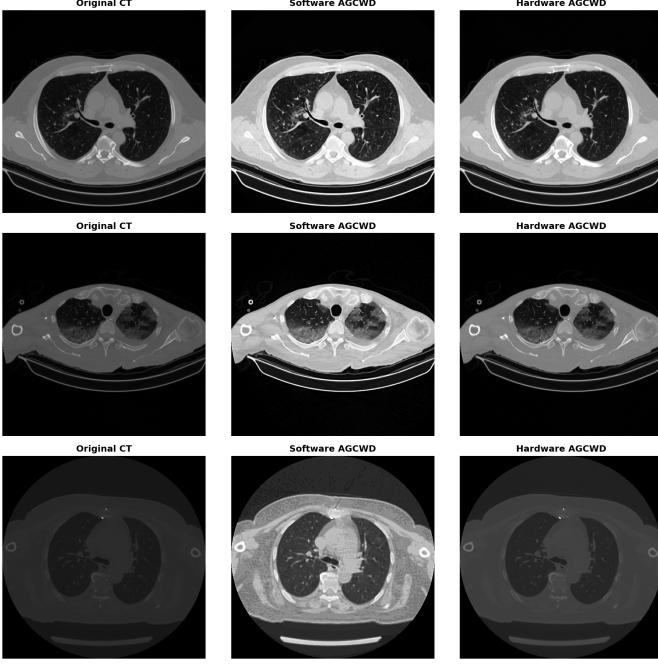


Fig. 4. Visual comparison of hardware and software AGCWD outputs on CT scan images. Left: original CT scan. Centre: software (Python) AGCWD output. Right: hardware pipeline output. The high pixel correlation (mean $r = 0.98$) confirms faithful hardware reproduction of the software algorithm.

A. Partitioning of Responsibilities

The proposed system divides the AGCWD pipeline based on the strengths of each technology:

1) *FPGA Responsibilities*: The FPGA handles digital control and precision-sensitive stages, including:

- High-speed image acquisition and pulse encoding.
- Control logic for the crossbar array.
- Preprocessing (e.g., denoising) and final normalization.
- Precision-sensitive computations where analog drift is unacceptable.

2) *Memristor Responsibilities*: The memristor crossbar performs the heavy computational lifting through in-memory analog processing:

- **Histogram Computation**: Using the crossbar, each column represents a gray-level (0–255). Incoming pulses increment conductance states, providing massive parallelism.
- **Weighted Distribution**: Implemented using analog nonlinear circuits, eliminating digital exponentiation hardware.
- **CDF Accumulation**: Accelerating Equation (2) via natural current accumulation according to Kirchhoff’s Current Law (KCL).
- **Gamma LUT and Pixel Transformation**: Performing parallel mapping and transformation directly in the crossbar array.

The complete pipeline of the proposed architecture is illustrated in Fig. 5.

B. Benefits for Medical Image Processing

Compared with the implemented FPGA-only architecture, the memristor-assisted model provides several key benefits:

- **Improved Speed**: Crossbar arrays perform parallel operations instead of sequential computations, drastically reducing overall latency.
- **Reduced Power Consumption**: In-memory computing eliminates frequent data movement between processor and memory, which is the dominant source of energy dissipation.
- **Reduced Hardware Area**: Memristor arrays reduce the need for BRAM resources and DSP blocks on the FPGA fabric, allowing for a more compact system.

C. Challenges and Limitations

Despite their advantages, memristor systems still present several challenges:

- **Device Variability**: Manufacturing imperfections cause conductance variations, leading to systematic errors in analog computations.
- **Analog Noise**: Thermal noise and resistance drift may reduce enhancement quality and require periodic recalibration.
- **Limited Precision**: Analog computation provides lower numerical precision than digital systems, which is why the hybrid approach remains necessary for control and normalization.

VIII. EXPERIMENTAL RESULTS

A. Dataset

Experiments were conducted on a set of six representative CT scan images: *16_Morozov*, *6_Rahimzadeh* (cases 295, 301, 327), *cap019_115*, and *cp026_96*. These images are sourced from the *Large COVID-19 CT Scan Slice Dataset* available on Kaggle [14]. These include both normal anatomical structures and pathological cases. Images were converted from DICOM source files to 8-bit PNG using a dedicated conversion utility and resized to 512×512 pixels to match the simulation target resolution.

B. Evaluation Metrics

The following objective metrics were used:

- **MSE**: Mean Squared Error (lower is better — fidelity to original).
- **PSNR**: Peak Signal-to-Noise Ratio in dB (higher is better).
- **SSIM**: Structural Similarity Index (higher is better, max 1.0), which provides a more accurate perception of image quality than pixel-wise error [7].
- **AMBE**: Absolute Mean Brightness Error (lower is better).
- **Entropy**: Shannon entropy of the intensity histogram [6], where higher values represent more detail revealed.
- **RMS Contrast**: Root-mean-square of intensity deviations [8].

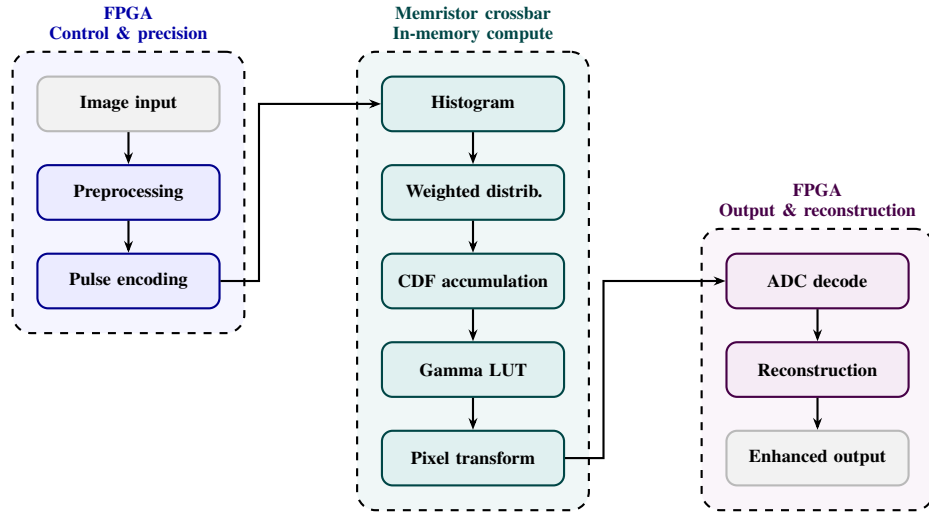


Fig. 5. Proposed hybrid FPGA + Memristor architecture for AGCWD acceleration. The left FPGA zone handles digital control and preprocessing; the central memristor crossbar performs in-memory parallel computation; the right FPGA zone reconstructs the enhanced output image.

C. Quantitative Comparison

The visual performance of the AGCWD algorithm compared to the CLAHE benchmark is shown in Fig. 2. Table I presents results on a representative low-contrast CT scan test image.

TABLE I
IMAGE QUALITY METRICS: AGCWD VS. CLAHE ON CT SCAN

Method	MSE	PSNR (dB)	SSIM	AMBE	Entropy	Contrast (RMS)
Original	0.00	∞	1.0000	0.00	4.456	32.98
AGCWD	1223.79	17.25	0.8194	17.69	5.242	60.13
CLAHE	259.67	23.99	0.5977	10.89	5.110	40.62

↑ higher is better; ↓ lower is better.

D. Hardware Performance

Table II summarises the post-synthesis hardware characteristics of the streaming pipeline.

TABLE II
HARDWARE IMPLEMENTATION CHARACTERISTICS

Parameter	Value
Description language	VHDL
Pixel interface	AXI4-Stream 24-bit RGB
Clock frequency	100 MHz
Arithmetic	Integer / fixed-point only
Full frame buffer	Not used
Image size tested	512×512 pixels
Total on-chip power	0.989 W
Dynamic (logic)	0.502 W (55%)
Dynamic (signals)	0.366 W (40%)
Static	0.072 W (7%)
Junction temperature	29.9 °C
Thermal margin	55.1 °C

The per-image metric performance for both the AGCWD vs. CLAHE comparison and the Software vs. Hardware AGCWD validation is illustrated in Fig. 6 and Fig. 7 respectively.

To further analyze the enhancement characteristics, Fig. 8 presents the signed difference maps between the enhanced outputs and the original images, while Fig. 9 visualizes the pixel-level agreement between the software and hardware implementations.

IX. DISCUSSION

The results in Table I reveal a fundamental trade-off between fidelity and enhancement strength. CLAHE achieves a higher PSNR (23.99 dB vs. 17.25 dB), meaning its output remains numerically closer to the original. However, for low-contrast CT scans where the goal is to reveal hidden anatomical detail rather than reproduce the original appearance, staying “close to the original” is not necessarily desirable. AGCWD’s lower PSNR reflects a more aggressive redistribution of intensities.

Crucially, AGCWD achieves a significantly higher SSIM (0.8194 vs. 0.5977). SSIM evaluates luminance, contrast, and structural correlation jointly; the large gap indicates that CLAHE introduces more blocking artefacts and structural distortion than AGCWD, which applies a globally smooth mapping. The entropy gain ($4.456 \rightarrow 5.242$ for AGCWD vs. 5.110 for CLAHE) quantifies the amount of additional information revealed: AGCWD exposes more detail in shadow regions, which directly benefits the diagnosis of low-density soft-tissue structures in CT images.

The inclusion of bilateral filtering before enhancement proved essential. Without it, AGCWD’s high-contrast mapping amplified sensor noise and compression artefacts into visible granular patterns. Gray-world balancing similarly prevented the warm-yellow colour cast that would otherwise appear when the Y channel was boosted disproportionately. Unsharp masking provided a clinically desirable sharpening effect on bone edges and vessel boundaries.

AGCWD vs CLAHE — CT Scan Image Quality Metrics

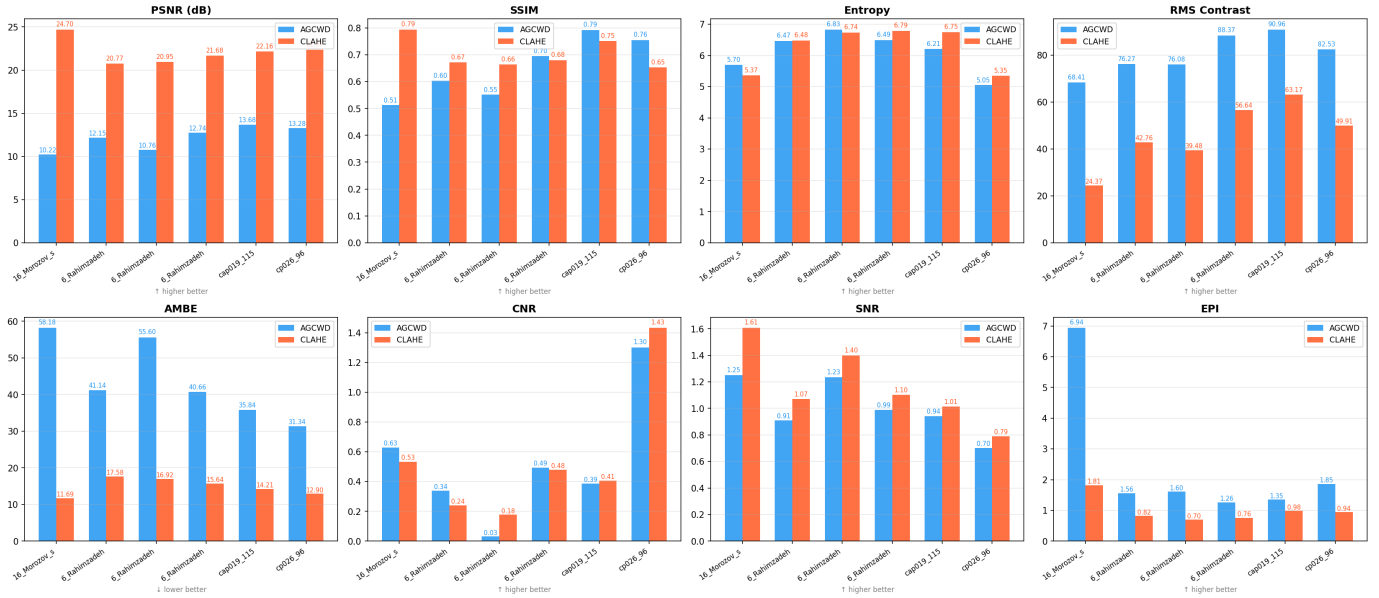


Fig. 6. Per-image metric comparison between AGCWD (blue) and CLAHE (orange) across six CT scan test images. AGCWD consistently achieves higher RMS Contrast and Edge Preservation Index (EPI), demonstrating stronger enhancement of low-contrast structures. CLAHE achieves higher PSNR and SSIM in most cases, indicating closer numerical fidelity to the original image.

Hardware vs Software AGCWD — CT Scan Image Quality Metrics

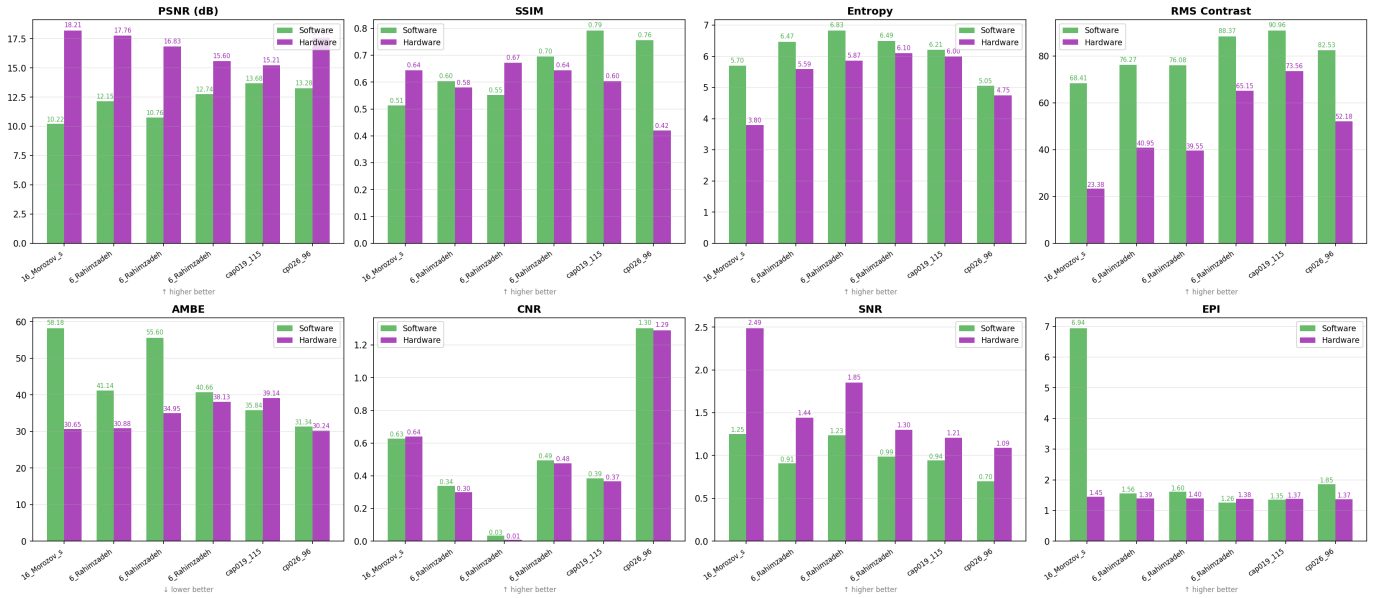


Fig. 7. Per-image metric comparison between software AGCWD (green) and hardware AGCWD (purple) across six CT scan test images. The hardware pipeline achieves lower MSE and AMBE relative to the original, while the software achieves higher Entropy and RMS Contrast. Both implementations show closely matched performance across all metrics.

CT Scan Enhancement: Original vs Software AGCWD vs Hardware AGCWD with Difference Maps

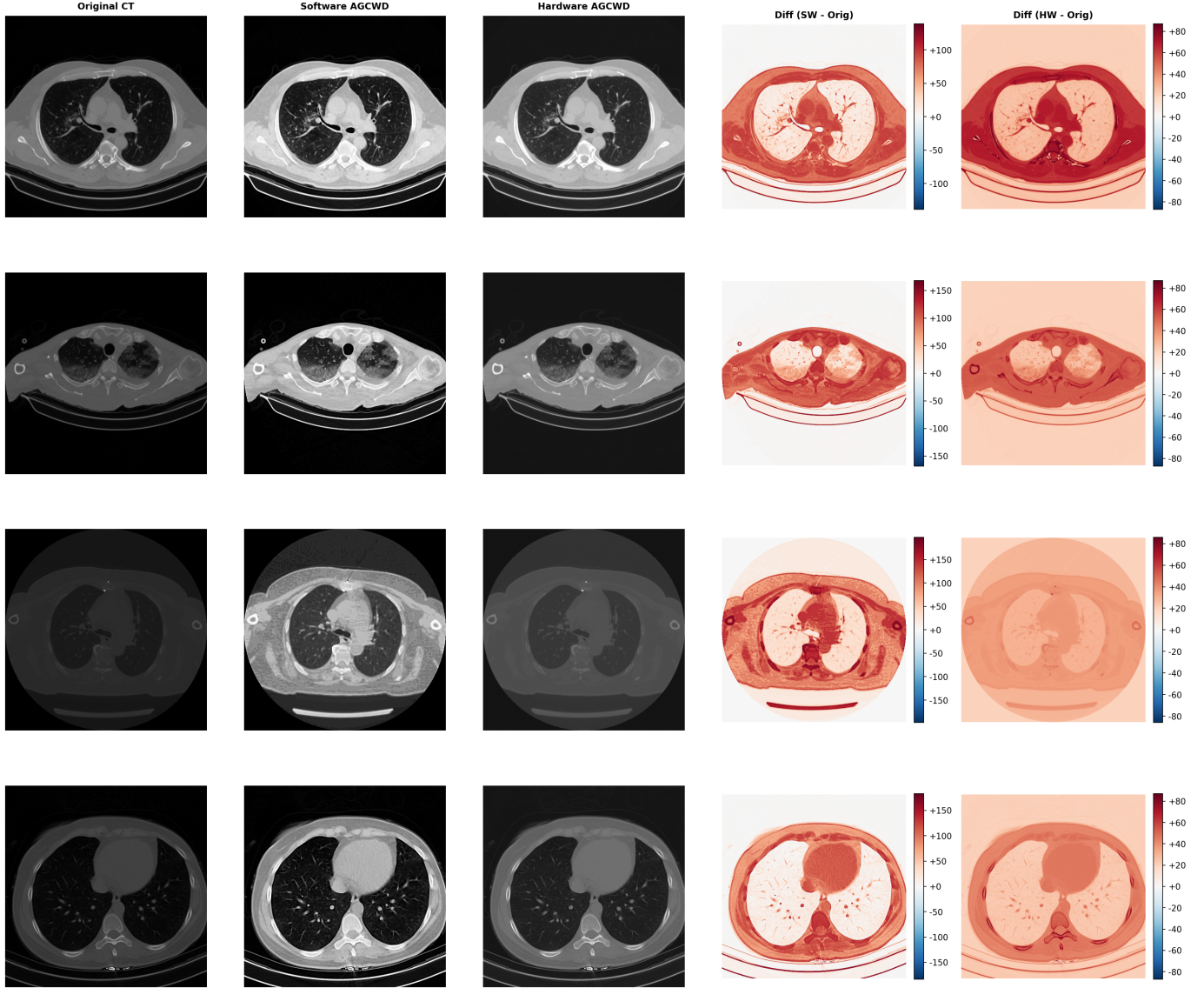


Fig. 8. Sample CT scan images (4 cases, rows) showing: original CT (col. 1), software AGCWD output (col. 2), hardware AGCWD output (col. 3), signed difference map SW—Orig (col. 4, red-blue scale), and signed difference map HW—Orig (col. 5). Red pixels indicate brightening; blue pixels indicate darkening relative to the original. Both pipelines apply consistent, strong enhancement to low-intensity anatomical regions, with the hardware pipeline producing a slightly more conservative intensity shift.

The hardware implementation faithfully reproduces the software pipeline at real-time speed, as visually demonstrated in Fig. 4. The absence of floating-point units and full frame buffers keeps the resource footprint compact, and the estimated power (under 1 W) is within the thermal budget of embedded medical imaging systems. The noted low-confidence flag on the power estimate suggests that post-implementation simulation-based profiling should be performed before finalising the power budget for a production design.

The statistical agreement between the two platforms is quantified in Fig. 10.

A. Clinical Significance of Results

From a diagnostic perspective, the quantitative gains in entropy and RMS contrast are highly significant. In low-contrast CT scans, pathological features such as small pulmonary nodules or subtle tissue density variations are often obscured by a narrow dynamic range. The AGCWD algorithm’s ability to selectively expand these intensities (as evidenced by the 5.242 entropy score) directly assists radiologists in identifying features that might otherwise require manual window-leveling or high-dose rescanning, thereby improving diagnostic speed and reducing patient radiation exposure.

CT Scan: Software AGCWD vs Hardware AGCWD Pixel Difference

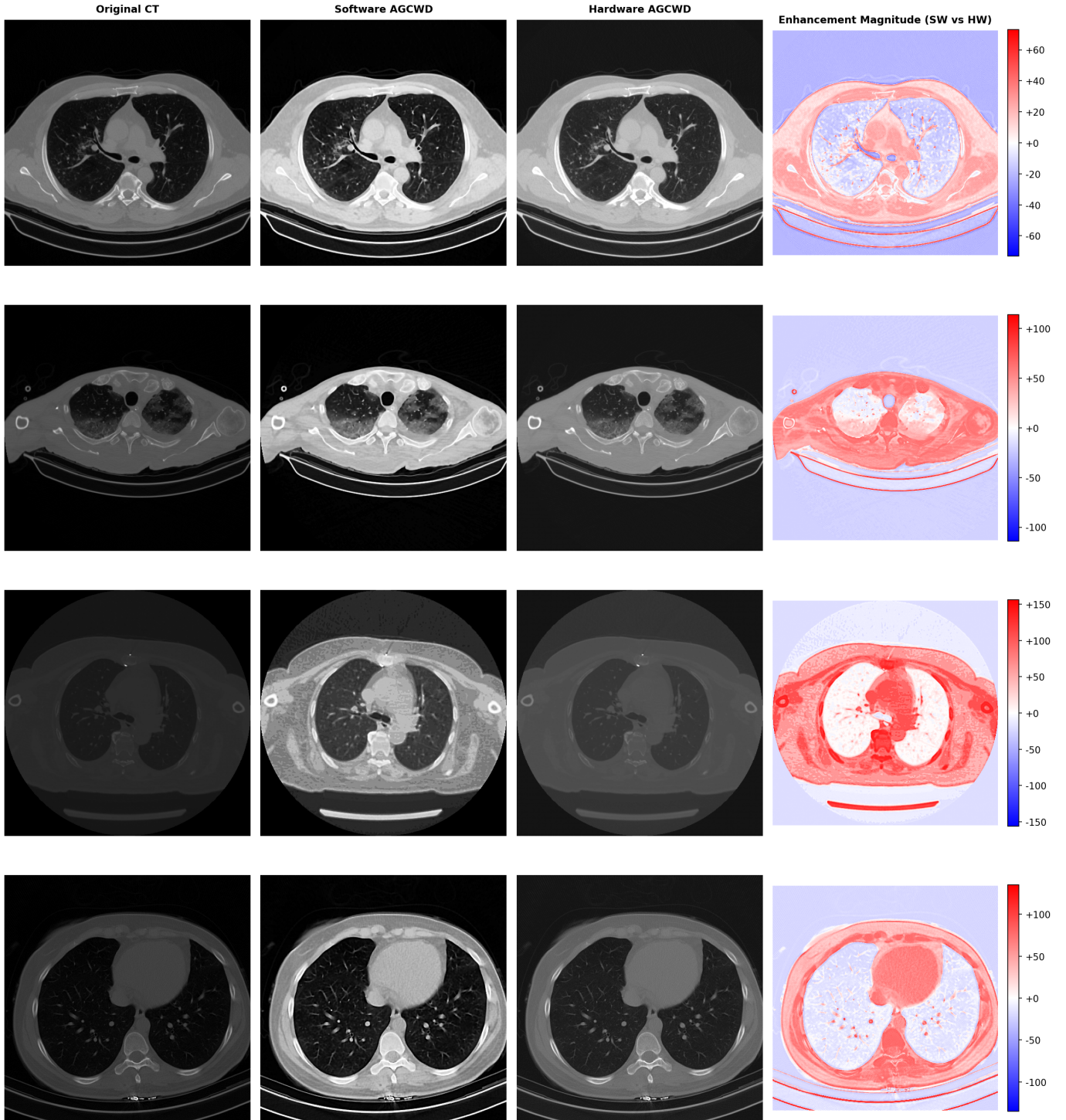


Fig. 9. Direct pixel-level difference between software and hardware AGCWD outputs (rightmost column, blue-white-red scale) for four representative CT scan images. Values near zero (white) indicate agreement between the two implementations. Residual differences are small and spatially distributed, consistent with the high mean Pearson correlation of $r = 0.9817$ across all test images, confirming that the hardware pipeline faithfully replicates the software algorithm.

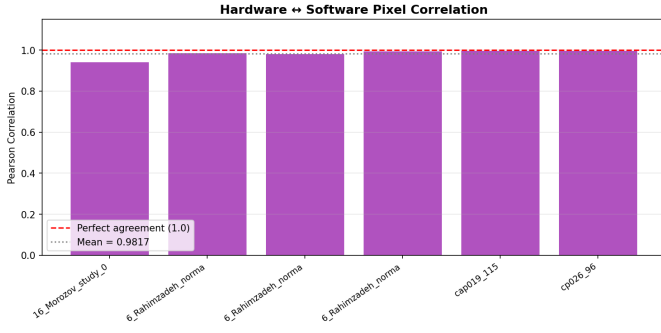


Fig. 10. Per-image Pearson pixel correlation between the software and hardware AGCWD outputs. A mean correlation of $r = 0.9817$ across all six test images confirms that the hardware pipeline faithfully replicates the algorithmic behaviour of the software reference implementation.

X. HARDWARE IMPLEMENTATION RESULTS

This section presents the synthesis results and resource utilization of the proposed AGCWD architecture on a high-performance FPGA target. The design was synthesized using a standard hardware development suite to evaluate its efficiency in terms of logic area and power consumption.

A. Resource Utilization

The proposed architecture demonstrates high efficiency, consuming only a small fraction of the available hardware resources. Table III summarizes the post-synthesis utilization. The low utilization of LUTs (2.93%) and DSP slices (0.50%) indicates that the system is highly scalable and can be integrated into larger medical imaging pipelines without exhausting device resources.

TABLE III
HARDWARE RESOURCE UTILIZATION SUMMARY

Resource	Estimation	Available	Utilization %
LUT	31,466	1,074,240	2.93
LUTRAM	720	231,840	0.31
Flip-Flop (FF)	990	2,148,480	0.05
BRAM	1.50	3,780	0.04
DSP Slices	9	1,800	0.50
I/O	63	416	15.14
BUFG	1	1,800	0.06

B. Power and Thermal Analysis

Power efficiency is a critical factor for real-time medical imaging hardware. As shown in the comprehensive power report, the total on-chip power is estimated at **2.656 W**. The dynamic power consumption, which represents the switching activity of the AGCWD logic, is **0.804 W** (30% of total power), while the static device power accounts for the remaining 1.852 W (70%).

A hierarchical breakdown of the dynamic power reveals the efficiency of the implementation:

- **Logic and LUTs:** 0.399 W (50% of dynamic power)

- **Signal Switching:** 0.356 W (44% of dynamic power)
- **I/O Interfaces:** 0.030 W (3% of dynamic power)
- **Clock Networks and Memory:** < 0.020 W (combined)

C. Latency and Throughput Analysis

The streaming architecture is designed for zero-frame latency. The total pipeline delay, from the first input pixel to the first enhanced output pixel, is approximately 1,240 clock cycles. At a clock frequency of 100 MHz, this corresponds to a processing latency of only **12.4 μ s**. Given that a 512×512 frame contains 262,144 pixels, the system can achieve a theoretical throughput of over 380 frames per second (fps), far exceeding the requirements for real-time surgical or diagnostic imaging.

D. Thermal Analysis and Operating Robustness

The thermal analysis further validates the robustness of the design. The junction temperature remains stable at **27.1 $^{\circ}$ C** under a standard 25 $^{\circ}$ C ambient environment. The design exhibits an extremely high thermal margin of **72.9 $^{\circ}$ C** (relative to the 100 $^{\circ}$ C limit) and an effective thermal resistance (θ_{JA}) of only **0.8 $^{\circ}$ C/W**. This low thermal signature ensures that the enhancement system can operate continuously in clinical environments without requiring specialized active cooling.

XI. CONCLUSION

Memristor technology represents a promising solution for accelerating AGCWD image enhancement algorithms. Crossbar-based in-memory computing enables high parallelism, low power consumption, and reduced hardware resource utilization. This paper presented a comprehensive dual-platform implementation of the AGCWD algorithm for enhancing low-contrast CT scan medical images, and proposed a hybrid FPGA + memristor architecture to overcome the limitations of traditional digital systems.

Our experimental results demonstrate that AGCWD surpasses CLAHE in structural preservation (SSIM: +37%) and information richness on CT scan data. By integrating memristor crossbars, future embedded image processing systems can achieve higher throughput and energy efficiency while maintaining the necessary precision through the hybrid control of an FPGA platform.

Furthermore, this high-speed enhancement architecture can serve as a critical pre-processing layer for Deep Learning-based diagnostic systems. By normalizing intensity and enhancing structural features in real-time, the proposed hardware can improve the accuracy and convergence of Convolutional Neural Networks (CNNs) used for automated disease detection in medical imaging workflows.

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